
**Robert Jenlu Lo, Ph.D.**
**National Central University**
**jeniuluo@gmail.com**


- **20+ years of hands-on experience**
- EUV / VUV Optical Systems Specialist
- UHV & Precision Metrology
- Automation & Data Processing
- AI/ML Integration

I am an R&D Builder and System Integration Architect with 20+ years of hands-on experience in building advanced optical systems (EUV/VUV) and ultra-high vacuum (UHV) physical experimental stations from 0 to 1.While my academic track record includes 70+ top-tier SCI journal publications (950+ citations, h-index 18), my core distinction lies in system-level cross-disciplinary troubleshooting. I bridge the gap between high-level theoretical physics and practical engineering execution to help companies reduce costs, increase yield, and break through critical R&D bottlenecks.

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|---|---|
| Personal Information | Male, 1974 born |
| Military Service | Completed (Mar 2000) |
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| Address | Zhubei City, Hsinchu County, Taiwan |
| Preferred Contact Method | E-mail, Mobile |
| Driver's Licenses | Medium/Heavy Truck<br>Heavy Motorcycle |

## Education:

| National Central University | National Taiwan University | National Chiao Tung University |
|---|---|---|
| Ph.D 2010 | Physics 1998 * | Civil 1992* |

* Discontinued

## Core Competencies Key Highlights

- Hardware R&D & System Integration
- Instrumentation, Automation & Data Analytics
- IT, Cloud & AI Infrastructure
- Precision Machining & Design for Manufacturability (DFM)
- Project Management & Team Leadership

## References

**Bing Ming Cheng**
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# Experience


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[Core Positioning]
- Full-Stack Physical & Software-Hardware Systems Architect
- Master Troubleshooter for Complex Equipment & Software-Hardware Bottlenecks

## Senior Independent Software & Hardware Integration Engineer
### Independent R&D Team (Research, 3 – 7 employees)
### Full-Stack Engineer, Cloud Architect, Systems Integration Consultant | Hsinchu City

2023/7~ now

- Built a personal AI server using Ollama: Satisfies strict privacy and data security requirements with zero subscription fees and low long-term maintenance costs, while offering high customizability, flexibility, offline availability, and open-source ecosystem integration.
- Deployed virtual machines (VMs) on GCP (Google Cloud Platform) and AWS (Amazon Web Services).
- Set up and operated rAthena RO servers.

GCP, AWS, AI, Cloud, VM, LLM, MySQL, MariaDB, rAthena

## Postdoctoral Research Fellow
### Hualien Tzu Chi Hospital (Educational Research, 500+ employees)
### Researcher | Hsinchu City

2019/12~2023/7
3years 8 months

- VUV, EUV, and X-ray imaging ellipsometry
- Applied domain expertise and research techniques to the medical field
- Collaborated with Academia Sinica on fluorescent diamond applications
- Wrote SQL queries in MariaDB and MySQL
- Implemented machine learning in Python

#Research Report & Paper Writing #Laboratory Equipment Operation

## Postdoctoral Research Fellow
### National Synchrotron Radiation Research Center (Research, 100 – 500 employees)
### Physics Researcher | Hsinchu City

2011/4~2019/11
8years 8 months

- Scientific research and paper publication (see attachment for details).
- R&D and system integration:
  - Measurement system for temperature-dependent VUV fluorescence and absorption spectra.
  - Filter system for suppressing high-coherence synchrotron radiation.
  - Photochemical research system: A unified platform capable of simultaneous absorption and fluorescence spectrum measurements across UV, visible, and IR regions.
- Integrated computer instrumentation and control (LabVIEW) for all above systems.
- Data analysis software development: Authored Origin scripts to reduce manual workload and enhance data processing speed and precision.
- Set up a Linux-based quantum chemistry server supporting parallel computing (Gaussian).

## Postdoctoral Researcher
### University of Southern California (Higher Education, 500+ employees)
### Physics Researcher | Hsinchu City

2010/7~2010/12
6 months

- Scientific research
- System R&D and integration:
  - Extreme ultraviolet (EUV) fluorescence spectroscopy system for gaseous samples with temperature variation measurement capabilities
  - Absorption spectroscopy and excitation spectroscopy measurement systems
- Paper publication and writing (see attachment for details)

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# Core Competencies & Key Highlights

[Core Positioning]
- Full-Stack Physical & Software-Hardware Systems Architect
- Master Troubleshooter for Complex Equipment & Software-Hardware Bottlenecks

## Hardware R&D & System Integration

- ✓ - Optics & Spectral Analysis: Expertise in Ultra-High Vacuum (UHV) optical system design; hands-on experience in R&D, assembly, and alignment of spectrometers/monochromators across X-ray, EUV, VUV, and IR spectra (fluorescence, absorption, and excitation spectroscopy).
- ✓ - Advanced Vacuum & High-Temperature Tech: Proficient in designing and assembling UHV systems (down to 10^-9 Torr); experienced in glow discharge/wall-stabilized arc lamp light sources, differential pumping systems, and high-temperature vacuum furnaces (1200˚C).
- ✓ - High Voltage & Particle Optics: Capable of high-voltage DC field control (up to 50 kV/cm^2) and insulation leakage suppression; experienced with ion sources (30 keV) and mass spectrometers (Quadrupole / TOF-MS).
- ✓ - Synchrotron Beamline Station Construction: Deep understanding of synchrotron beamline principles; proven track record in hardware/software modification, maintenance, and upgrades for beamlines and experimental stations (Hatches).

## Instrumentation, Automation & Data Analytics

- ✓ - Industrial Instrumentation & Automation (LabVIEW): Developed Human-Machine Interfaces (HMI) for multiple experimental systems, integrating automated Data Acquisition (DAQ) with synchronized beamline monochromator control.
- ✓ - Automated Data Processing & Modeling: Utilized OriginPro Script / Python for large-scale batch data cleaning and high-precision automated analysis; built physical/chemical theoretical models and simulations using Wolfram Mathematica.
- ✓ - Quantum Chemical Computation & HPC: Set up Linux (CentOS/Ubuntu) High-Performance Computing (HPC) parallel servers for molecular structure and energy level simulations using Gaussian 09.

## IT, Cloud & AI Infrastructure

- ✓ - Cloud & DevOps: Experienced in setting up GCP and AWS VM environments, Linux/Windows system maintenance, Samba data synchronization, and hosting web services (Web/Mail/phpBB).
- ✓ - Database Management: Proficient in MySQL and MariaDB database design, SQL scripting, and automated monitoring scripts.
- ✓ - Local AI & LLM Deployment: Skilled in deploying on-premise private AI servers via Ollama to achieve offline LLM application with zero subscription cost and high data privacy.
- ✓ - Programming Languages: Python, C/C++, C#, Visual Basic .NET, VBA, FORTRAN, Shell Script, HTML/PHP/JavaScript.

## Precision Machining & Design for Manufacturability (DFM)

- ✓ - CAD 3D Mechanical Design: Proficient in SolidWorks for 3D CAD design, engineering drawings, and system rendering.
- ✓ - Hands-on Machining & Maintenance (DFM Practitioner): Practical experience operating lathes, milling machines, EDM, electric welding, and TIG welding. Implements DFM principles during design to significantly reduce manufacturing complexity and cost.

## Project Management & Team Leadership

- ✓ - Research & Publication Output: Authored 70+ SCI international journal papers over the past decade, demonstrating exceptional logical reasoning and grant proposal writing skills.
- ✓ - Team Coordination & Cross-Domain Communication: Strong technical mentorship and cross-functional leadership skills that optimize workflows, boost team morale, and enhance operational efficiency.

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# Summary

[Core Positioning]
- Full-Stack Physical & Software-Hardware Systems Architect
- Master Troubleshooter for Complex Equipment & Software-Hardware Bottlenecks

I am proficient in mathematical and logical analysis, and able to apply these skills to any field, including solving various problems, designing and constructing new systems, programming experimental procedures in data acquiring, analyzing data, integrating software and hardware, improving system automation, and making systems more user-friendly.

In my past work experience, I have successfully designed and constructed various equipment and systems, including vacuum ultraviolet/ultraviolet light sources, atomic beam systems, time-of-flight mass spectrometers, spectrometers, Stark effect photoionization systems, vacuum ultraviolet absorption and fluorescence systems, vacuum ultraviolet photochemistry research systems, and more. At the "Vacuum Ultraviolet Absorption, Excitation, and Fluorescence Research Station" on beamline 03, and the "Ultra-high Flux Vacuum Ultraviolet Photochemistry Station" on beamline 21A2 at the National Synchrotron Radiation Center, I designed modifications to make them more automated and user-friendly, resulting in significant improvements in research accuracy and efficiency.

In the field of computer instrument control, I personally wrote all instrument control programs in the systems I developed, including the program that comunicates with synchrotron radiation spectrometer. Based on clear logic and physical knowledge, and in combination with the characteristics of computer instrument control and data analysis, I successfully broke through the limitations of the instrument and analyzed phenomena that were previously impossible to obtain. I also wrote OriginPro scripts to process large amounts of redundant data, significantly reducing labor costs and errors. I built theoretical models and wrote Wolfram Mathematica scripts to establish and simulate models. Additionally, I set up a CentOS Linux server with parallel computing capabilities, installed and configured Gaussian 09 on it, and wrote shell scripts to facilitate the use of colleagues and perform molecular energy level simulations.

I also set up a data storage system on Ubuntu system, utilizing the features of Samba server to connect data between Windows and Linux for synchronized backup. I also utilized the Google Cloud Platform to deploy database systems (MariaDB and/or MySQL), and set up web, mail, and phpBB servers. On this virtual machine, I also deployed an MMORPG game server, connected to the SQL server, and wrote SQL scripts to monitor database changes and maintain its stability.

In each experiment, various problems were encountered, and I was able to propose effective and quick solutions based on the aforementioned software and hardware capabilities. I also demonstrated strong communication skills, the ability to organize team members, and led by example to improve team cohesion. With everyone working together wholeheartedly, we were ultimately able to achieve impressive results in each experiment

I have the ability to quickly grasp the essence of technology, integrate related technologies, and find the most suitable solutions for various technological problems. With a broad and in-depth knowledge and skills, I am eager for opportunities in the industry to showcase my strengths and receive valuable rewards.


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# The Three Core Questions:

[Core Positioning]
- Full-Stack Physical & Software-Hardware Systems Architect
- Master Troubleshooter for Complex Equipment & Software-Hardware Bottlenecks

## Discontinued Studies at Two Universities

### From Rote-Learning Confusion to First Principles

✓ *Early in my academic journey, I experienced a period of exploration. My time in Civil Engineering at NCTU helped me realize that rote formula memorization was not my passion. Re-entering university to study Physics at NTU allowed me to discover my true passion: understanding how the world operates through first principles. Driven by curiosity, I took demanding courses in Electrical Engineering and Computer Science (Computer Architecture, Data Structures). Although traditional exam formats prioritizing syntax memorization posed challenges at the time, this experience grounded me in fundamental physics and laid a solid foundation in cross-disciplinary software and hardware integration. As proven in today's AI era, syntax memorization can be handled by tools, whereas* ***the ability to see through low-level system architectures and rapidly integrate software and hardware*** *remains my most valuable asset.*

## Eight Years of Master's and Doctoral Studies

### building comprehensive system engineering capabilities and technical leadership.

✓ *For me, those 8 years were an intensive masterclass in* ***hands-on craftsmanship, top-tier physics, and cross-disciplinary collaboration.*** *Building on practical machining skills from my upbringing, I brought hands-on execution into national-level laboratories—building UHV chambers, 1200℃ high-temperature furnaces, repairing spectrometers, modeling in SolidWorks, and scripting automated LabVIEW/Python tools. More importantly, I served as the technical backbone for interdisciplinary teams (mechanical and chemical engineering), translating physical principles into actionable guidance for junior researchers. Beyond constructing complex experimental systems, I developed* ***system leadership that solves technical bottlenecks while empowering the team to deliver results.***

## All Postdoctoral Roles

### An Easily Satisfied Disposition Combined with Fanatical Passion as a Chief System Architect

✓ *Although my title was Postdoctoral Researcher, my actual function was* ***Chief System Architect*** *for top-tier beamlines at NSRRC. Upon joining, I fully automated previously manual and error-prone processes—building a 12-window vacuum filter chamber, rewriting LabVIEW control programs, and creating Origin data-processing templates. These upgrades enabled our team to publish dozens of SCI papers efficiently. I remained in academia because it provided the freedom to solve extreme technical challenges. Today, I am transitioning to the industry because I recognize that businesses need battle-tested system architects* ***who can apply first-principles thinking to solve highly complex software-hardware integration problems.***

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# Personality Strengths

[Core Positioning]
- Full-Stack Physical & Software-Hardware Systems Architect
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## Physics-School Geek combined with an Elite Hands-on Craftsman

### "First-Principles" Driven System Integration

✓ **Traits** You avoid rote memorization and seek a deep understanding of underlying logic (from calculus and electromagnetism to self-learning C/Python and replacing capacitors in mass spectrometers). You can instantly identify software/hardware bottlenecks and tame complex equipment using intuitive, cost-effective methods (e.g., off-the-shelf standard parts).

### Systems Engineering Vision (EPICS Full-Stack Implementation Capability)

✓ **Traits** Possess comprehensive experience in building experimental stations from scratch (0 to 1), including system design, hardware-software integration, and automated deployment. Capable of seamlessly integrating optics, vacuum systems, cryogenics, instrumentation & control, data pipelines, and HMIs into a stable, closed-loop operating system, thereby realizing a maintainable and scalable architecture for long-term operations.

### Full-Stack Hands-On Capability Backed by Craftsmanship and Instinct

✓ **Traits** Proficient in machining (turning, milling, welding), 3D CAD modeling (SolidWorks), vacuum systems, and optics. Capable of developing LabVIEW/Python automation and configuring Linux servers. Long-term practical experience has honed your refined sensory instincts (e.g., detecting pump anomalies by ear, distinguishing metals by color).

### Selfless Mentorship & "Technical Anchor" Leadership

✓ **Traits** Naturally inclined to assist others. When cross-disciplinary teammates (mechanical/chemical engineering) or visiting foreign scholars hit technical dead ends, you guide them step-by-step using solid logic and patience. You lead not through authority, but through technical problem-solving power, earning trust as a reliable mentor.

### Extreme Focus & Problem-Solving Mission Drive

✓ **Traits** An "On-Off" mental switch when encountering technical challenges. Once switched on, you work relentlessly around the clock to overcome hurdles (e.g., working late into the night or building complex systems within a week).